# Lab W2D4: GPU Image

✅ Step 1 — Create Dockerfile.gpu

```
(base) n00n1@Nadia:~/serving-stack$ cd ~/serving-stack
(base) n00n1@Nadia:~/serving-stack$ nano Dockerfile.gpu
(base) n00n1@Nadia:~/serving-stack$ ls
 Dockerfile        Dockerfile.naive                                          README.md   evidence    verify.sh
 Dockerfile.gpu   'Lab W2D3 Containerise '$'\342\200\223'' Evidence Report.pdf'   app         verify.py   verify_cell.py
(base) n00n1@Nadia:~/serving-stack$ cat Dockerfile.gpu
FROM nvidia/cuda:12.4.1-runtime-ubuntu22.04

RUN apt-get update && apt-get install -y --no-install-recommends \
      python3.11 python3-pip python3.11-venv && \
    rm -rf /var/lib/apt/lists/* && \
    ln -sf /usr/bin/python3.11 /usr/local/bin/python && \
    ln -sf /usr/bin/python3.11 /usr/local/bin/python3

WORKDIR /app

COPY app/requirements.txt .

RUN pip3 install --no-cache-dir -r requirements.txt

COPY app/ .

EXPOSE 8000

CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
(base) n00n1@Nadia:~/serving-stack$
```

✅ Step 2 — Build GPU Image (Completed)

```
(base) n00n1@Nadia:~/serving-stack$ docker build -f Dockerfile.gpu -t Nadiax94/serving-stack:gpu-v1 .
[+] Building 1345.5s (11/11) FINISHED                                                                  docker:default
 => [internal] load build definition from Dockerfile.gpu                                                         0.0s
 => => transferring dockerfile: 544B                                                                              0.0s
 => [internal] load metadata for docker.io/nvidia/cuda:12.4.1-runtime-ubuntu22.04                                0.2s
 => [internal] load .dockerignore                                                                                 0.0s
 => => transferring context: 101B                                                                                 0.0s
 => [1/6] FROM docker.io/nvidia/cuda:12.4.1-runtime-ubuntu22.04@sha256:517da2300c184c9999ec203c2665244bdebd3578d1  0.2s
 => => resolve docker.io/nvidia/cuda:12.4.1-runtime-ubuntu22.04@sha256:517da2300c184c9999ec203c2665244bdebd3578d1  0.0s
 => [internal] load build context                                                                                 0.0s
 => => transferring context: 2.93kB                                                                               0.0s
 => [2/6] RUN apt-get update && apt-get install -y --no-install-recommends       python3.11 python3-pip python3.  33.8s
 => [3/6] WORKDIR /app                                                                                            0.0s
 => [4/6] COPY app/requirements.txt .                                                                             0.0s
 => [5/6] RUN pip3 install --no-cache-dir -r requirements.txt                                                  1098.3s
 => [6/6] COPY app/ .                                                                                             0.2s
 => exporting to image                                                                                          208.2s
 => => exporting layers                                                                                         156.4s
 => => exporting manifest sha256:b6878b718030eb4e53ec2e508e324384fd5646ea804702c2220266f8aa95148f                 0.1s
 => => exporting config sha256:02bd10acbef2eeae8a795ab95975e764147cbb8661f4c76ad4cf16a7945a0817                   0.0s
 => => exporting attestation manifest sha256:2fc842c8960b8a6d06c1d9a0a2cdafe0f17c72d301ceb2913a6e1eee2da0852e     0.0s
 => => exporting manifest list sha256:34f42387714e9bc86e32bda55743625ce7de6e3bd34c8bf57e61ab615d304bc1            0.0s
 => => naming to Nadiax94/serving-stack:gpu-v1                                                                    0.0s
 => => unpacking to Nadiax94/serving-stack:gpu-v1                                                                51.5s
(base) n00n1@Nadia:~/serving-stack$ docker images
                                                                                              i Info →  U In Use
IMAGE                                            ID             DISK USAGE   CONTENT SIZE   EXTRA
Nadiax94/serving-stack:gpu-v1                    34f42387714e       12.6GB          4.6GB
aidc-serving:naive                               631f89019983       16.5GB          6.6GB
ghcr.io/mohnadbabgi/aidc-16-pythonserver:latest  52d6e727271f       2.13GB          539MB
ghcr.io/nadiax94/aidc-10-warmup:latest           dcc78266c0bc       2.13GB          539MB
hello-world:latest                               5dd0d3e6e255       25.9kB         9.49kB    U
n00n1/aidc-serving:cpu-v1                        a3af1442c1bd       1.61GB          336MB    U
nadiax/aidc-serving:cpu-v1                       a3af1442c1bd       1.61GB          336MB    U
nvidia/cuda:12.4.1-runtime-ubuntu22.04           517da2300c18       3.77GB         1.47GB
python:3.11-slim                                 9c900dea9e8f        189MB         47.7MB
sagemath/sagemath:latest                         e068670ae586       4.76GB         1.46GB    U
team-server-model:latest                         76756b95b2be       2.16GB          569MB
team-server:latest                               ed842ceafdd8        187MB         45.5MB
(base) n00n1@Nadia:~/serving-stack$
```

✅ Step 3 — Run GPU image and CPU fallback

```
(base) n00n1@Nadia:~/serving-stack$ docker run -d \
--name serving-gpu \
-p 8000:8000 \
-v hf-cache:/home/app/.cache/huggingface \
Nadiax94/serving-stack:gpu-v1
4948d23ea49624db302a12fc76bd493dd2214e8fa90050d04c276dd51dea60c0
(base) n00n1@Nadia:~/serving-stack$ docker ps
CONTAINER ID   IMAGE                            COMMAND                  CREATED              STATUS              PORTS
                                     NAMES
4948d23ea496   Nadiax94/serving-stack:gpu-v1   "/opt/nvidia/nvidia_…"   About a minute ago   Up About a minute   0.0.0.0
:8000->8000/tcp, [::]:8000->8000/tcp   serving-gpu
```

GPU image running with CPU fallback

```
(base) n00n1@Nadia:~/serving-stack$ docker logs -f serving-gpu

==========
== CUDA ==
==========

CUDA Version 12.4.1

Container image Copyright (c) 2016-2023, NVIDIA CORPORATION & AFFILIATES. All rights reserved.

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A copy of this license is made available in this container at /NGC-DL-CONTAINER-LICENSE for your convenience.

WARNING: The NVIDIA Driver was not detected.  GPU functionality will not be available.
   Use the NVIDIA Container Toolkit to start this container with GPU support; see
   https://docs.nvidia.com/datacenter/cloud-native/ .

INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

✅ Step 4 CPU fallback health check

```
(base) n00n1@Nadia:~$ curl http://localhost:8000/health
{"status":"ok","model":"Qwen/Qwen2.5-0.5B-Instruct"}(base) n00n1@Nadia:~$
```

✅ Step 5 — Docker Hub login success

```
(base) n00n1@Nadia:~/serving-stack$ docker login
Authenticating with existing credentials... [Username: nadiax]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```

✅ Step 6 — Docker image pushed successfully

```
(base) n00n1@Nadia:~/serving-stack$ docker push nadiax/serving-stack:gpu-v1
The push refers to repository [docker.io/nadiax/serving-stack]
6cd34cfd6491: Pushed
44136fa355b3: Mounted from nadiax/aidc-serving
0d6448aff889: Mounted from nvidia/cuda
3c645031de29: Mounted from nvidia/cuda
0a7674e3e8fe: Mounted from nvidia/cuda
b71b637b97c5: Mounted from nvidia/cuda
56dc85502937: Mounted from nvidia/cuda
ec6d5f6c9ed9: Mounted from nvidia/cuda
47b8539d532f: Mounted from nvidia/cuda
fd9cc1ad8dee: Mounted from nvidia/cuda
83525caeeb35: Mounted from nvidia/cuda
d386cb194d02: Pushed
7f0828d543d8: Pushed
e85591321c29: Pushed
e3b0dd114b8e: Pushed
cd2261083680: Pushed
gpu-v1: digest: sha256:34f42387714e9bc86e32bda55743625ce7de6e3bd34c8bf57e61ab615d304bc1 size: 856
(base) n00n1@Nadia:~/serving-stack$
```

Docker image tag verification

```
(base) n00n1@Nadia:~/serving-stack$ docker images | grep serving-stack
Nadiax94/serving-stack:gpu-v1                    34f42387714e       12.6GB          4.6GB    U
nadiax/serving-stack:gpu-v1                      34f42387714e       12.6GB          4.6GB    U
```

✅ Step7 — Colab Tesla T4 GPU detected

```
[1] !nvidia-smi
0s

Wed Aug 26 04:26:19 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.82.07              Driver Version: 580.82.07      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   42C    P8              9W /   70W |       0MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

✅ Step 8 — Colab CUDA probe result

```
[4]
!find / -name generate_probe.py 2>/dev/null

/generate_probe.py
```

```
[5]
!python /generate_probe.py

device: cuda (Tesla T4), dtype: torch.float16
tokenizer_config.json: 7.30kB [00:00, 26.2MB/s]
vocab.json: 2.78MB [00:00, 80.7MB/s]
merges.txt: 1.67MB [00:00, 135MB/s]
tokenizer.json: 7.03MB [00:00, 177MB/s]
config.json: 100% 660/660 [00:00<00:00, 5.64MB/s]
2026-08-26 04:45:13.445959: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use availabl
To enable the following instructions: AVX2 AVX512F FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
model.safetensors: 100% 3.09G/3.09G [00:33<00:00, 92.7MB/s]
generation_config.json: 100% 242/242 [00:00<00:00, 2.58MB/s]
The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may ob
/usr/local/lib/python3.13/dist-packages/transformers/generation/configuration_utils.py:590: UserWarning: `do_sample` is set to `Fals
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/transformers/generation/configuration_utils.py:595: UserWarning: `do_sample` is set to `Fals
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/transformers/generation/configuration_utils.py:612: UserWarning: `do_sample` is set to `Fals
  warnings.warn(
{
  "cuda": true,
  "device_name": "Tesla T4",
  "tokens_per_s": 25.5,
  "model": "Qwen/Qwen2.5-1.5B-Instruct"
}
wrote gpu_evidence.json
```

```
[6]
!ls /

bin                      home     NGC-DL-CONTAINER-LICENSE  srv
boot                     kaggle   opt                       sys
content                  lib      proc                      tmp
cuda-keyring_1.1-1_all.deb lib32  python-apt                tools
datalab                  lib64    python-apt.tar.xz         usr
dev                      libx32   root                      var
etc                      media    run
generate_probe.py        mnt      sbin
```

## ✅ Step 9 — Final Green Check PASS

```
(base) n00n1@Nadia:~/serving-stack$ IMAGE=nadiax/serving-stack:gpu-v1 ./verify.sh
resolving nadiax/serving-stack:gpu-v1 (pull, else expect a local build) ...
waiting for /health on CPU fallback (up to 420s) ...
OK:Tesla T4:25.5
part 1: GPU image resolved
part 2: /health 200 on CPU fallback
part 3: colab evidence shows cuda: true
GREEN CHECK: PASS
```